Exercises I : Topology III

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1. Let X and Y be CW complexes, and $X * Y$ be the join of X and Y. Show that we obtain a CW complex structure on $X * Y$ which contains X and Y as subcomplexes. Also, in this CW complex structure

$$\#\{n\text{-cells of } X * Y\} = \#\{n\text{-cells of } X\} + \#\{n\text{-cells of } Y\} + \sum_{i+j=n-1} \#\{i\text{-cells of } X\} \cdot \#\{j\text{-cells of } Y\}.$$

2. Note that $S^0 * X \cong S(X)$, the unreduced suspension of X. Show that this has a CW complex structure with

$$\#\{0\text{-cells of } S(X)\} = \#\{0\text{-cells of } X\} + 2,$$

$$\#\{n\text{-cells of } S(X)\} = 2\#\{(n-1)\text{-cells of } X\} + \#\{n\text{-cells of } X\} \text{ for } n \geq 1.$$

3. Let S^1 be the space with C_m-action by multiplication by m^{th}-roots of unity. Demonstrate a CW complex structure with m 0-cells and m 1-cells, on which C_m acts by permuting the cells. Show that the resultant chain complex may be written as (where $C_m = \langle g \mid g^m = 1 \rangle$)

$$C_k^{CW}(S^1) = 0 \text{ if } k \geq 2, \quad C_1^{CW}(S^1) \to C_0^{CW}(S^1) \equiv \mathbb{Z}[C_m] \stackrel{1-g}{\to} \mathbb{Z}[C_m].$$

4. Using the above, show that there is a CW complex structure on S^{2n+1} with S^{2n-1} as a subcomplex, and having m $2n$-cells and m $2n+1$-cells. Consider the usual action of C_m on $S^{2n+1} = S(\mathbb{C}^{n+1})$ via multiplication by m^{th}-roots of unity. Show that S^{2n+1} has a CW complex structure with m-cells in each degree between 0 and $2n+1$. Further observe that C_m acts by permuting the cells. Show that the resulting cellular chain complex is equivalent to the following complex of length $2n+1$

$$\mathbb{Z}[C_m] \stackrel{g-1}{\to} \mathbb{Z}[C_m] \stackrel{N}{\to} \mathbb{Z}[C_m] \stackrel{g-1}{\to} \mathbb{Z}[C_m] \cdots \mathbb{Z}[C_m] \stackrel{N}{\to} \mathbb{Z}[C_m] \stackrel{g-1}{\to} \mathbb{Z}[C_m],$$

where $N = 1 + g + g^2 + \cdots + g^{m-1}$.

5. Consider the Lens space $L^{2n+1}(C_m) = S^{2n+1}/C_m$. Show that $L^{2n+1}(C_m)$ has a CW complex structure with one cell in each degree between 0 and $2n+1$. Using the $\mathbb{Z}[C_m]$-module structure on $C_*^{CW}(S^{2n+1})$ as above, show that

$$C_*^{CW}(L^{2n+1}(C_m)) \cong C_*^{CW}(S^{2n+1}) \otimes_{\mathbb{Z}[C_m]} \mathbb{Z}.$$

Compute $H_*(L^{2n+1}(C_m))$ and $H_*(L^{2n+1}(C_m); \mathbb{Z}/m)$. Also compute $H^*(L^{2n+1}(C_m))$ and $H^*(L^{2n+1}(C_m); \mathbb{Z}/m)$ (the additive structure only).

6. Consider the map $S^{2n+1}/C_m \to S^{2n+1}/S^1 = \mathbb{C}P^n$. Compute the induced map $H^2(\mathbb{C}P^n) \to H^2(L^{2n+1}(C_m))$.